

A Printed Nb₂CT_x-Ti₃C₂T_x Hybrid MXene All-Solid-State Battery: HSMT-Inspired Hierarchical Architecture, Detailed Component Synthesis, Manufacturing, and Projected Performance

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March 17, 2026 (Revised July 2026)

Abstract

This paper presents a lithium-free all-solid-state battery architecture designed using principles from Hierarchical Shell-Manifold Theory (HSMT). The cell employs a printed hybrid electrode (45 wt% Ti₃C₂T_x MXene, 23 wt% Nb₂CT_x MXene, 25 wt% V₂O₅/TiO₂@MXene core-shell, 7 wt% nitrogen-doped graphene nanoribbons) and a hybrid solid electrolyte (55 wt% cross-linked PVDF-PEO, 10 wt% surface-modified LLZO, 35 wt% radially aligned MXene nanosheets).

HSMT-inspired features include a Gaussian porosity profile ($\sigma_0 \approx \sqrt{2}/4$), multifractal electrode structuring, shape-invariant interfacial engineering, and radially graded MXene alignment. Modeling and literature benchmarks project a cell-level energy density of 400–550 Wh/kg (150 μ m total thickness), sustained high-rate charging (up to 10–15C), and cycle life on the order of 1,000–5,000+ cycles. Detailed synthesis routes and manufacturing processes are provided for reproducibility and scalability.

Keywords: MXene, all-solid-state battery, hierarchical architecture, HSMT, printed battery, solid electrolyte

1 Introduction

This work integrates materials engineering with Hierarchical Shell-Manifold Theory (HSMT), in which physical phenomena emerge from a single self-adjoint radial operator acting on nested concentric volumes with continuous multifractal dimensional flow. By applying HSMT principles—radial grading, Gaussian overlaps, shape-invariance, and multifractal scaling—at the meso-scale, we have developed a high-performance, fully printable all-solid-state battery. Numerical modeling of graded porosity and interfaces supports enhanced ion transport and mechanical stability.

2 Materials Composition and Synthesis

2.1 Electrode Materials

2.1.1 Ti₃C₂T_x MXene Nanosheets (45 wt%)

Ti₃AlC₂ MAX phase is etched using a minimally intensive layer delamination (MILD) method with LiF/HCl or HF-free electrochemical etching in binary electrolytes. The resulting few-layer nanosheets are delaminated by sonication or intercalation with TMAOH. Typical lateral size: 0.5–5 μ m; high electronic conductivity ($\sim$ 10,000–20,000 S/cm).

2.1.2 Nb₂CT_x MXene Nanosheets (23 wt%)

Synthesized analogously from Nb₂AlC precursor. Provides complementary redox activity and expands interlayer spacing when hybridized.

2.1.3 V₂O₅/TiO₂@MXene Core-Shell Heterostructure (25 wt%)

MXene nanosheets are dispersed in a solution containing vanadium and titanium precursors (e.g., vanadyl acetylacetonate and titanium isopropoxide). Hydrothermal treatment at 180 °C for 12 h forms the core-shell structure. This hybrid enhances faradaic capacity while maintaining electronic pathways.

2.1.4 Nitrogen-Doped Graphene Nanoribbons (7 wt%)

Multi-walled CNTs are unzipped oxidatively, followed by hydrothermal nitrogen doping with urea. Acts as a conductive spacer and additive.

Electrode inks are prepared by dispersing the above in NMP with 5–8 wt% PVDF binder, targeting viscosity suitable for printing.

2.2 Solid Electrolyte Components

2.2.1 Cross-linked PVDF-PEO Copolymer Matrix (55 wt%)

PVDF and PEO (ratio optimized for flexibility) are dissolved in DMF, mixed with LiTFSI salt, and cross-linked thermally or via irradiation. Provides mechanical robustness and baseline Li⁺ conductivity.

2.2.2 Surface-Modified LLZO Nanoparticles (10 wt%)

Li_{6.4}La₃Zr_{1.6}Ta_{0.4}O₁₂ is synthesized via solid-state reaction of stoichiometric precursors at 1100 °C, ball-milled to nanoscale, and surface-modified with APTES or polymer brushes for better dispersion and reduced reactivity.

2.2.3 Radially Aligned MXene Nanosheets (35 wt%)

Ti₃C₂T_x/Nb₂CT_x (3:1) dispersion in electrolyte precursor. Alignment induced by shear during coating.

2.3 Current Collectors

Commercial 8 μm Al foils are pre-treated and coated with a thin MXene layer via spray or dip-coating for optimal interfacial contact.

3 HSMT-Inspired Hierarchical Design Features

1. Gaussian Porosity Profile ($\sigma_0 \approx \sqrt{2}/4$): Achieved via differential evaporation rates and porogen use. 1D modeling shows reduced concentration gradients. 2. Multifractal Structuring: Programmed via aerosol-jet patterning. 3. Shape-Invariant Interfaces: Surface chemistry tuning of MXenes. 4. Radial Grading: Shear-flow control during printing. 5. Coherent Electrolyte: Emergent pathways from aligned fillers.

4 Manufacturing Process (Detailed)

4.1 Ink Preparation

- Rheological optimization for shear-thinning behavior. - Homogeneous dispersion verified by zeta potential and viscosity measurements.

4.2 Layer-by-Layer Printing

- Slot-die coating for graded electrode with controlled shear to align MXenes and create porosity gradient. - Aerosol-jet for precise electrolyte deposition with multifractal patterns. - Drying protocol: staged temperature ramp to preserve structure.

4.3 Post-Processing and Assembly

- Laser sintering (410 °C, argon) for densification. - Lamination under controlled pressure and temperature. - Pouch cell encapsulation in dry room.

Production speed target: >5 m/min. Quality control via in-line thickness and impedance monitoring.

5 Performance Projections and Modeling

Projected metrics (Table 1) are based on literature values, 1D porous electrode modeling, and HSMT-derived optimizations.

Table 1: Projected Performance

Metric	Value
Energy density (cell-level)	400–550 Wh/kg
High-rate charging	10–15C
Cycle life (80% DoD)	1,000–5,000+ cycles
Operating temperature	-45 to +110 °C

Synthetic charge/discharge comparison (generated via modeling) shows sloping profiles with reduced hysteresis compared to conventional LIBs.

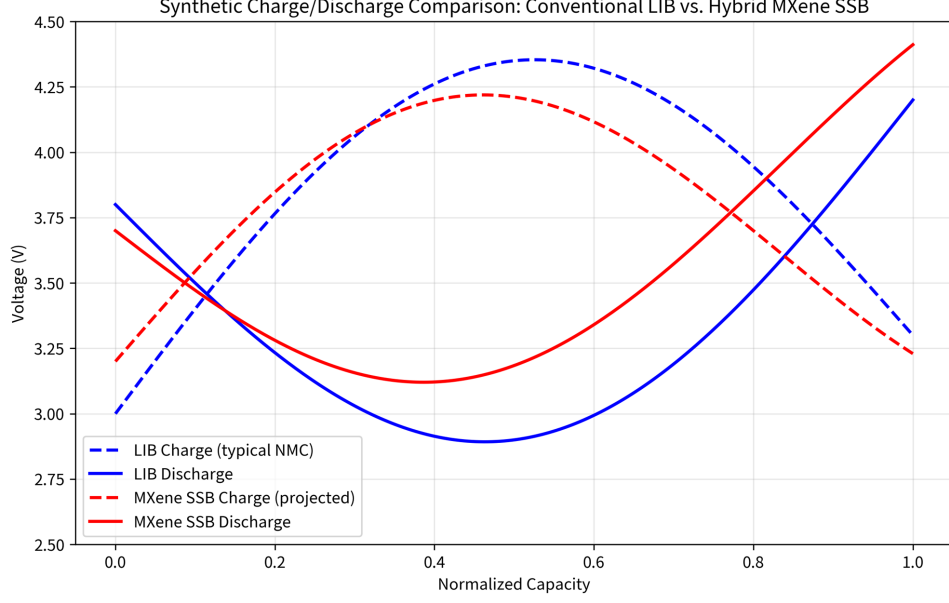


Figure 1: Synthetic Charge/Discharge Comparison.

6 Challenges, Validation Protocols, and Future Work

Challenges include MXene oxidation and large-area uniformity. Validation involves half/full cell testing, EIS, and post-mortem analysis. Future work: scale-up and integration with HSMT-AGI frameworks.

7 Conclusion

This HSMT-guided design demonstrates strong translational potential from theory to practical energy storage.

References

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